

MarginReach: Margin-Aware Residual Reinforcement Learning for Pre-Harvest Reaching in Deformable Fruit Canopies

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Abstract—Robotic fruit harvesting in dense, deformable canopies requires contact-rich motion through compliant vegetation. However, treating every branch as an obstacle makes planning overly conservative, while optimizing only reaching success can drive reinforcement learning policies to push through vegetation with excessive contact force or branch deflection. The central challenge is therefore to exploit mild branch compliance while reducing mechanically unsafe interaction. To make this trade-off explicit, we present MarginReach, a margin-aware residual reinforcement learning framework that converts contact force and branch deflection into normalized utilization signals, allowing the controller to adjust local Cartesian residual corrections according to the measured severity of canopy interaction during pre-harvest reaching. Our contributions are threefold. First, we introduce a normalized force/deflection utilization interface for safety-aware learning and evaluation, mapping arm-link contact forces and branch deflections to continuous robot-contact and plant-damage risk indicators. Second, we propose MarginReach, which augments a behavior-cloned reaching prior with reactive Cartesian residual corrections guided by contact and branch utilization signals. Third, we provide comprehensive simulation and real-robot evaluation protocols demonstrating that multi-modal margin utilization improves safe-success rates and execution robustness under randomized branch interference. The results indicate that margin-utilization feedback helps the learner balance nominal reaching success with mechanical safety, improving safe canopy interaction while reducing contact-force and branch-deflection overloads. Project website: <https://marginreach.github.io/MarginReach/>.

Index Terms—Robotics and Automation in Agriculture and Forestry, Reinforcement Learning, Imitation Learning, Robot Safety, Contact Modeling.

I. INTRODUCTION

ROBOTIC fruit harvesting sits at the critical intersection of structured robot manipulation and unstructured, highly dynamic agricultural environments [1], [2]. To successfully pick a fruit, a manipulator must routinely navigate through dense, deformable canopies where branches severely constrain the feasible approach directions [3]. In structured or heavily pruned orchards, classical perception-planning-grasping pipelines can effectively find collision-free paths [4], [5]. However, in wild or high-density canopies, treating every vegetative obstacle as a rigid, impenetrable hazard creates a severe mismatch between kinematic planning feasibility and practical reachability. In these complex scenarios, a strictly zero-contact path rarely exists. Conversely, gentle, deliberate contact with compliant branches is often inevitable and can be actively exploited to facilitate a safe, functional pre-harvest approach.

This operational reality exposes two critical research gaps in the current agricultural robotics literature. First, existing simulation and planning frameworks often oversimplify vegetation as static obstacles or report only binary collision outcomes. Even when compliant branches are modeled, they typically do not provide a normalized interface that relates contact force and branch deflection to robot-contact and plant-damage risk. Second, from a learning and control perspective, optimization objectives in deformable canopies remain highly underdeveloped. Standard reinforcement learning (RL) policies optimized solely for reaching success tend to aggressively exploit compliant simulation physics by violently pushing through branches, risking hardware damage. Conversely, pure imitation learning (IL) or behavior cloning (BC) policies remain brittle and prone to compounding tracking errors when unexpected branch resistance perturbs the demonstrated nominal path [6], [7]. Consequently, a fundamental paradigm shift is required: robot-environment interactions should not be treated as binary collision failures, but quantified through continuous force- and deflection-based indicators that expose how close the robot and surrounding branches are to their admissible mechanical limits.

To bridge these gaps, this paper introduces MarginReach, a margin-aware residual reinforcement learning framework that redefines physical contact as graded instantaneous utilization signals. Rather than relying on sparse, binary collision labels, our approach embeds compact contact and branch utilization signals directly into the interactive learning loop. The key insight is that minor, non-destructive branch deflection can be actively utilized to clear an approach corridor, while trajectories approaching critical link-wise force limits or joint-level branch deflection thresholds should receive stronger penalties before overload events occur.

Our complete framework uses a procedurally synthesized canopy as an interactive testbed for a structured hybrid controller. On top of standard L-system tree generation and spring-actuated branch models, we add normalized utilization readouts so that contact is represented as a graded risk signal rather than a binary event. Branch deflections provide joint-wise branch utilization, while simulated link-wise contact sensors provide contact-force utilization for the manipulator. Inside this interactive environment, MarginReach combines an inverse kinematics (IK)-based behavior-cloned prior with a residual reinforcement learning policy guided by the formulated margin utilization signals. The BC prior leverages offline demonstrations to supply a smooth, global nominal command toward the target region. Meanwhile, the residual network learns localized Cartesian corrections, dynamically adapting the end-effector trajectory to yield to or bypass unexpected

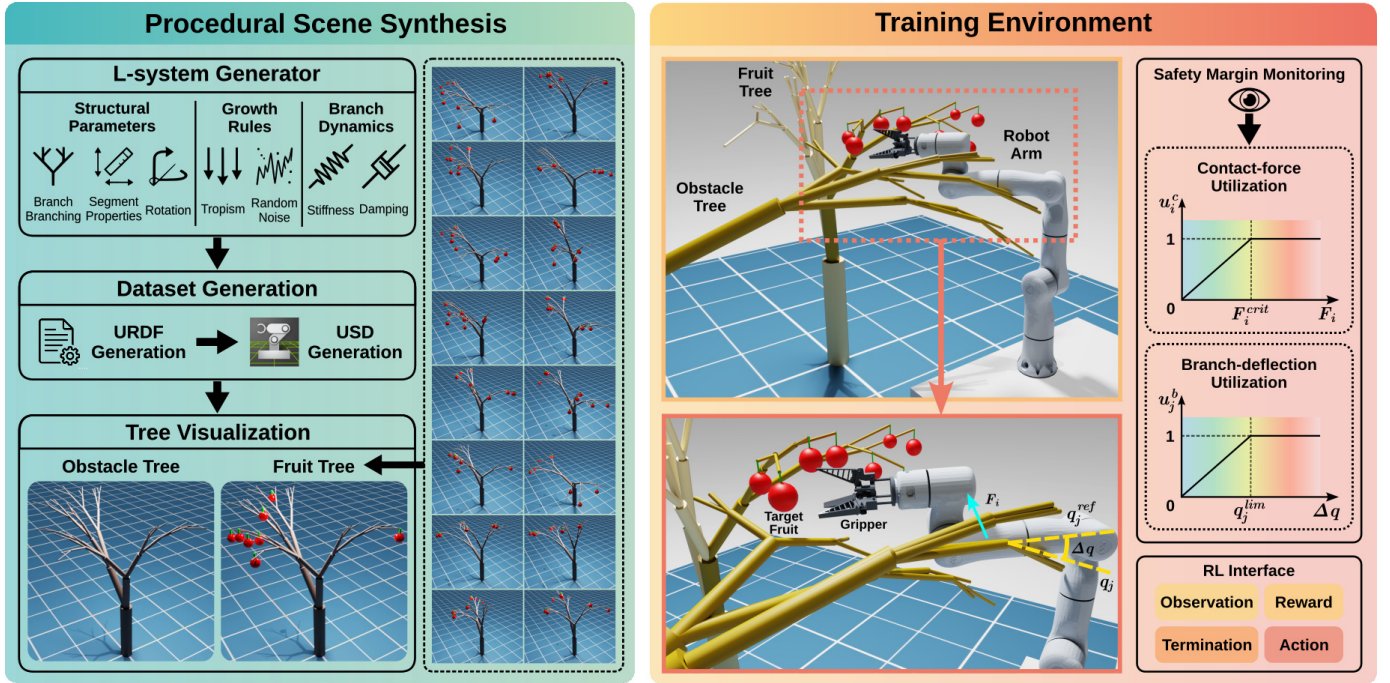


Fig. 1. Procedural canopy synthesis and training environment. Left: the L-system pipeline generates randomized fruit-tree and obstacle-tree assets with branch dynamics. Right: an Isaac Lab scene in which a Rizon arm reaches a target fruit through clutter from a separate obstacle tree.

branch resistance.

The primary contributions of this work are threefold:

- 1) A normalized force/deflection utilization interface for contact-rich pre-harvest fruit reaching, which converts manipulator contact forces and branch-joint deflections in a procedural Isaac Lab canopy into continuous indicators of robot-contact and plant-damage risk.
- 2) MarginReach, a margin-aware residual reinforcement learning framework that augments a behavior-cloned prior with reactive Cartesian corrections guided by continuous, link-wise contact and joint-wise branch utilization signals.
- 3) Comprehensive simulation and real-robot evaluation protocols demonstrating that the proposed residual learning framework, driven by multi-modal margin utilization signals, improves safe-success rates and reduces overload rates in our experiments under highly randomized branch interference.

II. RELATED WORK

A. Fruit Harvesting and Canopy Manipulation

Automated fruit harvesting has moved from isolated fruit detection and end-effector design toward interactive manipulation in cluttered foliage [1]–[3]. Traditional sampling-based planners generally model vegetation as static, rigid obstacles [4], [5], whereas recent canopy-manipulation methods explicitly model or exploit branch compliance through graph abstractions [8], articulated spring-damper models [9], or deliberate pushing and sliding strategies [10], [11]. These studies establish the importance of intentional branch interaction, but they mainly address structural modeling, contact detection, or open-loop trajectory modification. In contrast, MarginReach

exposes force and deflection as normalized utilization variables for closed-loop policy learning and evaluation.

B. Learning Contact-Rich Manipulation Policies

GPU-accelerated simulators have made model-free RL practical for contact-rich robot control [12]–[15], including recent fruit-harvesting policies with zero-shot sim-to-real transfer [6]. For branch interaction, Jacob *et al.* [7] introduced PCAP, a proprioceptive contact-aware policy that leverages whole-arm tactile and contact cues, and later whole-body exploration frameworks learned multi-link interactions within deformable clutter [16]. However, contact awareness alone does not guarantee structural or mechanical safety: binary collision or overload penalties provide sparse feedback only after unsafe events, while analytical Safe RL methods require safe sets that are difficult to define in stochastic botanical canopies [17], [18]. MarginReach instead uses simulator feedback to expose continuous contact and branch proximity-to-limit indicators before critical thresholds are breached.

C. Imitation and Residual Robot Control

Training model-free RL from scratch under contact and safety constraints is sample-inefficient and can require hazardous early exploration [15]. Imitation learning, behavior cloning (BC), and DAgger-style data aggregation mitigate this issue by initializing policies from expert demonstrations and relabeling visited states [19]–[21]. In our setting, an inverse-kinematics demonstrator supplies smooth nominal reaching trajectories but cannot cover off-path states induced by randomized branch deflections or force feedback. Residual control addresses this coverage deficit by retaining a prior controller while superimposing learned corrections [22]–[24]. MarginReach follows this prior-guided structure, but optimizes the

residual policy under simultaneous force and branch-deflection utilization signals rather than tracking or reach-time objectives alone.

III. PROCEDURAL PRE-HARVEST REACHING ENVIRONMENT

A. Tree Generation

As summarized in Fig. 1, the environment is built by generating randomized tree assets and instantiating them in Isaac Lab. Following the parametric L-system representation used for procedural plant generation in competitive manipulation frameworks [7], [25], we define a structural grammar $G = \langle V, \Sigma, \omega, P \rangle$, where V denotes the module alphabet, Σ is the parameter set, ω represents the axiom, and P is the set of production rules. We implement a sympodial growth branching grammar, where the turtle-graphics modules are directly interpreted as branch segment plotting and local frame rotation commands:

$$\begin{aligned} \Sigma_s : \{ & r_1 = 0.75, r_2 = 0.70, a_1 = 20^\circ, \\ & a_2 = 40^\circ, w_r = 1/\sqrt{3}, n = 6 \}, \\ \omega : & A(100, 10), \end{aligned}$$

$$p_A : A(\ell, w) \rightarrow !(w)F(\ell)[\mathcal{R}_L(a_1)B(\ell r_1, ww_r)] \\ \mathcal{R}_H(3\pi/4)[\mathcal{R}_L(a_2)B(\ell r_2, ww_r)], \quad (1)$$

$$p_B : B(\ell, w) \rightarrow !(w)F(\ell)[\mathcal{R}_U(a_1)\mathcal{H}B(\ell r_1, ww_r)] \\ [\mathcal{R}_U(-a_2)\mathcal{H}B(\ell r_2, ww_r)]. \quad (2)$$

Here, $F(\ell)$ advances the turtle state to instantiate a cylindrical branch segment of length ℓ , $!(w)$ specifies the local branch diameter, and the square brackets $[\cdot]$ push and pop the turtle state to form independent lateral sub-branches. The geometric operators $\mathcal{R}_L$, $\mathcal{R}_U$, and $\mathcal{R}_H$ perform spatial rotations about the turtle's left, up, and heading axes, respectively. The operator $\mathcal{H}$ acts as a horizontalization constraint that reorients the local left vector relative to the gravity vector before executing recursive growth.

To provide rich geometric and scene diversity, the nominal parameters specified in (1)–(2) are stochastically perturbed prior to each expansion sequence: the scaling factors r_1, r_2 are uniformly sampled within ± 0.15 , the branching angles a_1, a_2 within $\pm 10^\circ$, and the radius ratio w_r within ± 0.2 of their nominal values. During state interpretation, a downward tropism vector field vectorially biases each forward segment by slightly reorienting the turtle frame about $\mathbf{H} \times \mathbf{T}$, where $\mathbf{H}$ is the current heading vector and $\mathbf{T} = e[0, 0, -0.5]^\top$ represents a gravity-aligned attraction vector with intensity $e = 0.5$. The synthesized polyline skeleton is subsequently compiled into rigid cylindrical branch links within the unified robot description format (URDF) for Isaac Lab ingestion. For fruit-bearing tree instances, rigid spherical targets are appended to selected distal branch endpoints via a thin peduncle model consisting of two orthogonal, unactuated revolute joints. This universal-joint abstraction enables natural two-axis swinging behaviors under physical manipulation while maintaining a non-yielding attachment to the terminal branch.

B. Compliant Scene Model

Each simulation episode concurrently instantiates one randomized fruit-bearing target tree and one independently sampled obstacle tree. The target tree is systematically anchored within the reachable workspace in front of the robot manipulator, whereas the obstacle tree is randomly positioned along the approach corridor to introduce non-trivial branch clutter and multi-link physical interference.

To model vegetation compliance under external robot interaction, we employ a spring-actuated joint abstraction inspired by prior tree-manipulation models [7]. Each continuous L-system branch segment is represented as a rigid cylinder, and adjacent segments are interconnected by revolute joints that act as equivalent bending hinges. For a specific branch joint j featuring a parent segment length L_j and radius r_j , the equivalent bending stiffness and damping gains of the angular drive are configured as:

$$K_{p,j} = \frac{E\pi r_j^4}{2L_j}, \quad K_{d,j} = \frac{K_{p,j}}{10}, \quad (3)$$

where E denotes the effective Young's modulus of the vegetation material. The coefficient in (3) defines an effective hinge approximation that lumps local bending compliance into a single angular degree of freedom for the articulated branch model. These parameters are directly mapped to the native joint-level angular drives within the Isaac USD assets during runtime compilation.

To quantify branch-deflection risk during interaction in the physics engine, a nominal bending-limit angle q_j^{lim} is assigned to each active branch joint. Under the equivalent bending-hinge model, the limit is estimated from the outer-fiber bending strain of a circular branch segment. With small-curvature approximation $\kappa_j \approx q_j/L_j$, the maximum bending strain is $\epsilon_{\max,j} \approx r_j q_j/L_j$. Enforcing $E_j \epsilon_{\max,j} \leq \sigma_{\max,j}$ gives:

$$q_j^{lim} = \frac{\sigma_{\max,j} L_j}{E_j r_j} = \frac{2\sigma_{\max,j} L_j}{E_j d_j}, \quad (4)$$

where L_j is the branch segment length, r_j is the radius, $d_j = 2r_j$ represents the local branch diameter, E_j denotes the effective Young's modulus, and $\sigma_{\max,j}$ is the nominal allowable bending stress. The resulting q_j^{lim} serves as the reference boundary for normalizing the branch deflection utilization metric.

C. Margin Utilization

Unlike binary collision labels or raw force/deflection readings, margin utilization maps heterogeneous interaction signals onto a common normalized scale for safety-aware learning and evaluation. We formulate this metric as an instantaneous proximity-to-limit indicator for robot-contact force and branch deflection. Throughout this work, index i references a specific manipulator link, whereas index j indexes an environment branch joint.

For an individual robot link i , let F_i be the net contact force magnitude evaluated by the physics engine. Given a

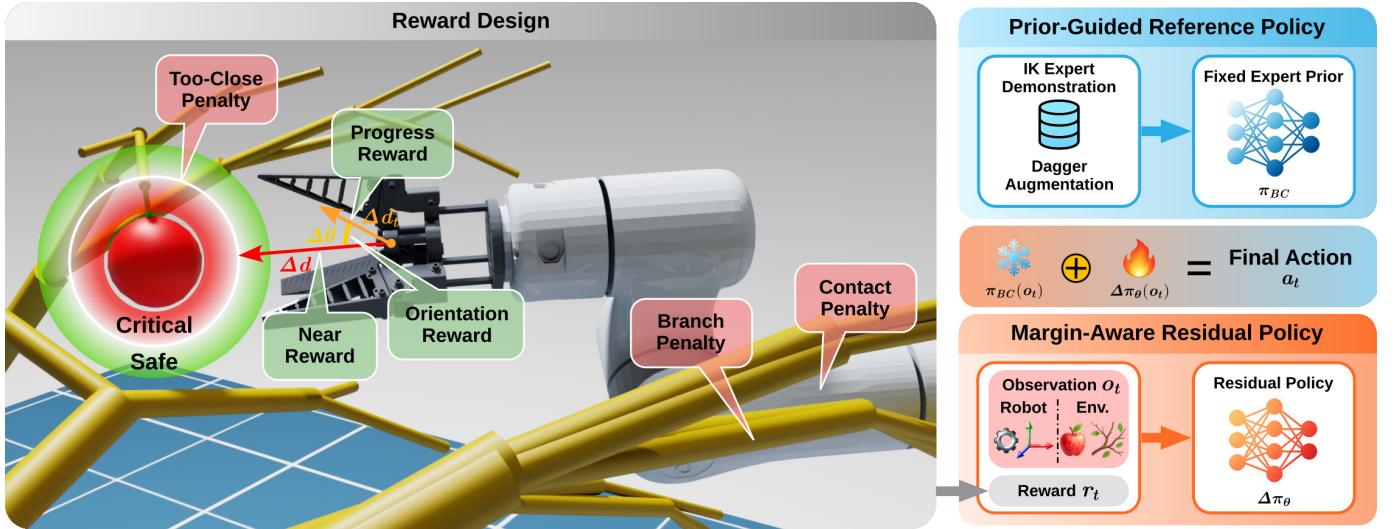


Fig. 2. Control and reward structure of MarginReach. A frozen BC/IK policy supplies the nominal Cartesian action, and PPO learns a residual correction. The objective jointly promotes pre-harvest reaching and penalizes contact-force and branch-deflection utilization.

predefined, link-specific critical force threshold F_i^{crit} , the continuous contact utilization u_i^c is mapped as:

$$u_i^c = \min\left(\frac{F_i}{F_i^{crit}}, 1\right). \quad (5)$$

Here, $u_i^c = 0$ signifies an uncontacted link state, values within the open interval $(0, 1)$ indicate increasing instantaneous proximity to the contact-force limit, and $u_i^c = 1$ indicates that the critical hardware force limit has been reached or breached.

Concurrently, for an environmental branch joint j , the branch deflection utilization u_j^b is defined relative to its passive equilibrium configuration. Upon episode initialization, the canopy physics scene is stepped until settling under gravity, and the passive resting angle of each joint is recorded as q_j^{ref} . The branch utilization u_j^b is then formulated as:

$$u_j^b = \min\left(\frac{|q_j - q_j^{ref}|}{q_j^{lim}}, 1\right), \quad (6)$$

where q_j^{lim} is the nominal bending-based deflection threshold from (4). Analogously, $u_j^b = 0$ indicates that the branch remains at its structural equilibrium, intermediate values denote increasing instantaneous proximity to the deflection threshold, and $u_j^b = 1$ indicates that the prescribed deflection limit has been reached or exceeded.

IV. MARGINREACH: MARGIN-AWARE RESIDUAL RL

A. Task Formulation

Given the procedural environment in Sec. III, MarginReach formulates pre-harvest fruit reaching as an MDP, with the residual policy and reward interface organized as shown in Fig. 2. The task is defined around reaching a pre-harvest standoff pose; nominal reaching success is therefore specified by the end-effector pose relative to the target fruit. Let $\mathbf{o}_t$ denote the policy observation:

$$\mathbf{o}_t = [\mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{x}_t^{ee}, \mathbf{p}_t^{fruit}, \mathbf{u}_t^c]. \quad (7)$$

Here $\mathbf{q}_t, \dot{\mathbf{q}}_t \in \mathbb{R}^7$ are the arm joint positions and velocities, $\mathbf{x}_t^{ee} \in \mathbb{R}^7$ is the end-effector position and quaternion relative to the robot base frame, and $\mathbf{p}_t^{fruit} \in \mathbb{R}^3$ is the relative target position. The contact utilization vector $\mathbf{u}_t^c \in \mathbb{R}^8$ contains seven arm-link utilizations from (5) and their maximum. Branch utilization is computed from simulator joint states and used for reward, termination, and evaluation.

A rollout is counted as a nominal reaching success when the end-effector reaches a desired pre-harvest standoff distance from the target fruit. Let d_t denote the end-effector–fruit distance. The distance condition is $|d_t - d_s| \leq \epsilon_d$, where d_s is the desired standoff distance and ϵ_d is a small tolerance; in the current configuration, $d_s = 10$ cm and $\epsilon_d = 1$ cm. The approach axis must also satisfy the orientation tolerance. Sec. V further evaluates whether successful rollouts also satisfy the contact-force and branch-deflection limits. The standoff distance places the end-effector at a favorable pre-harvest pose for subsequent gripper closure and detachment, while a separate too-close condition penalizes commands that drive through the fruit or stem.

The policy action $\mathbf{a}_t \in \mathbb{R}^6$ is a local Cartesian pose increment: the first three dimensions command translation and the last three command an axis-angle orientation update. This action is executed by a damped-least-squares IK controller,

$$\Delta \mathbf{q}_t = \mathbf{J}^\top (\mathbf{J}\mathbf{J}^\top + \lambda^2 \mathbf{I})^{-1} \Delta \mathbf{x}_t, \quad (8)$$

where $\mathbf{J}(\mathbf{q})$ is the manipulator Jacobian, λ is the damping factor, and $\Delta \mathbf{x}_t$ is the Cartesian pose increment induced by $\mathbf{a}_t$. Because residual RL commands can vary at high frequency near contact transitions, both the Cartesian pose increments and the resulting IK joint targets are smoothed using an exponential moving average, e.g., $\bar{\mathbf{x}}_t = \alpha \Delta \mathbf{x}_t + (1 - \alpha) \bar{\mathbf{x}}_{t-1}$. This low-pass filtering improves command continuity, reduces policy-induced oscillations, and mitigates force spikes when the arm interacts with compliant branches.

B. Reward Design

The reward first encourages the arm to approach the target fruit, then stabilizes it in the pre-harvest standoff band while discouraging unsafe contact. The band is the same one used in Sec. IV-A: $d_{min} = d_s - \epsilon_d$ and $d_{max} = d_s + \epsilon_d$.

The reaching reward (r_t^{reach}) has three parts. The progress term (r_t^{prog}) rewards reductions in the end-effector–fruit distance while the arm remains outside the standoff band. The near term (r_t^{near}) is centered at the desired standoff distance rather than at the fruit center. The orientation term (r_t^{ori}) is activated once the end-effector enters a neighborhood of the fruit, $d_t < d_{ori}$, so that pose alignment is shaped only near the target:

$$\begin{aligned} r_t^{prog} &= \mathbb{I}[d_t > d_{max}](d_{t-1} - d_t), \\ r_t^{near} &= \exp\left[-\left(\frac{d_t - d_s}{\sigma_d}\right)^2\right], \\ r_t^{ori} &= \mathbb{I}[d_t < d_{ori}]c_t^{ori}, \quad d_{ori} = \beta_{ori}d_s, \\ r_t^{reach} &= \omega_p r_t^{prog} + \omega_n r_t^{near} + \omega_o r_t^{ori}. \end{aligned} \quad (9)$$

Here σ_d controls the width of the standoff reward, $\beta_{ori} > 1$ sets the distance at which orientation shaping begins, and c_t^{ori} is the quaternion-alignment cosine between the end-effector and target orientations.

The risk penalty (p_t^{risk}) covers three unsafe behaviors. We use $\rho(x) = (\max(x, 0))^2$ to denote a squared positive-part function. The too-close penalty (p_t^{close}) is activated only when the end-effector moves inside the standoff band. The contact penalty (p_t^c) and branch penalties ($p_t^{b,*}$) directly reuse the clipped utilization variables defined in (5) and (6). Let N_c be the number of monitored arm links and N_b^* be the number of branch joints for the fruit tree ($*$) = f or obstacle tree ($*$) = o :

$$\begin{aligned} p_t^{close} &= \rho\left(\frac{d_{min} - d_t}{d_{max} - d_{min}}\right), \\ p_t^c &= \frac{1}{N_c} \sum_{i=1}^{N_c} (u_{i,t}^c)^2 + \lambda_c \left(\max_i u_{i,t}^c\right)^2, \\ p_t^{b,*} &= \frac{1}{N_b^*} \sum_{j=1}^{N_b^*} (u_{j,t}^{b,*})^2 + \lambda_b \left(\max_j u_{j,t}^{b,*}\right)^2, \quad * \in \{f, o\}, \\ p_t^{risk} &= \omega_{tc} p_t^{close} + \omega_c p_t^c + \omega_{bf} p_t^{b,f} + \omega_{bo} p_t^{b,o}. \end{aligned} \quad (10)$$

The averaged quadratic terms provide dense multi-channel shaping from all monitored links and branch joints, so that non-maximum contact or deflection channels are not ignored during learning. The maximum-utilization terms, weighted by larger coefficients, provide the primary overload-avoidance signal by penalizing the link or branch joint closest to its limit. Hard-limit violations are handled by the terminal failure indicator. To reduce fruit shaking during the pre-harvest approach, the fruit-side branch penalty $p_t^{b,f}$ also includes the utilization of the fruit-stem rotational joints. These reward and termination terms provide safety-aware shaping signals that reduce empirical threshold violations in the evaluated rollouts.

TABLE I
REWARD PARAMETERS.

Group	Symbols	Values
Terminal	ω_s, ω_f	80, 40
Reach	$\omega_p, \omega_n, \omega_o$	100, 5, 5
Orientation	$\beta_{ori} d_s$	0.25 m
Risk	$\omega_{tc}, \omega_c, \omega_{bf}, \omega_{bo}$	4, 3, 5, 5
Max utilization	λ_c, λ_b	10, 10
Regularization	$\omega_a, \omega_v, \omega_T$	0.005, 0.005, 0.02

The terminal reward (r_t^{term}) combines success and failure events as

$$r_t^{term} = \omega_s \mathbb{I}_t^{safe} - \omega_f \mathbb{I}_t^{fail}, \quad (11)$$

where $\mathbb{I}_t^{safe}$ is one only when four conditions hold together: the end-effector is inside $[d_{min}, d_{max}]$, the approach orientation is aligned, contact utilization is below its limit, and branch utilization is below its limit. The failure indicator $\mathbb{I}_t^{fail}$ covers excessive contact force, excessive branch deflection, or entering the too-close region.

The total reward (r_t) is then

$$r_t = r_t^{term} + r_t^{reach} - p_t^{risk} - \omega_a \|\mathbf{a}_t\|^2 - \omega_v \|\dot{\mathbf{q}}_t\|^2 - \omega_T, \quad (12)$$

where ω_T is a per-step time cost. The action, joint-velocity, and time penalties regularize the residual controller and discourage unnecessarily long episodes.

C. Prior-Guided PPO

We first construct a kinematic demonstration policy that generates local Cartesian reaching commands with DLS-IK. The resulting observation–action pairs are used to train a behavior-cloned reference policy π_{BC} by supervised action regression. In practice, the actor is optimized with a mean-squared imitation loss on normalized Cartesian actions, with observation noise augmentation and optional DAGger-style relabeling to improve robustness near obstacle-induced deviations [21]. The learned reference policy provides a smooth nominal approach to the pre-harvest band, and the residual policy complements it when incidental branch contact changes the feasible local motion.

During reinforcement learning, π_{BC} is kept fixed and PPO trains a residual Cartesian policy $\Delta\pi_\theta$ from the same observation $\mathbf{o}_t$. The executed Cartesian action is

$$\mathbf{a}_t = \pi_{BC}(\mathbf{o}_t) + \Delta\pi_\theta(\mathbf{o}_t), \quad (13)$$

where $\mathbf{o}_t$ is the margin-augmented observation in (7). The residual actor is initialized near zero correction so that training starts close to the imitation policy and then learns contact- and branch-aware adjustments under the reward in (12).

V. EXPERIMENTS

A. Training and Evaluation Setup

The simulation study uses the procedural environment in Sec. III and the margin-aware policy formulation in Sec. IV. Each scene contains one fruit-bearing target tree and one obstacle tree without fruit. Tree geometry, target-fruit selection, and obstacle-tree placement are randomized during training.

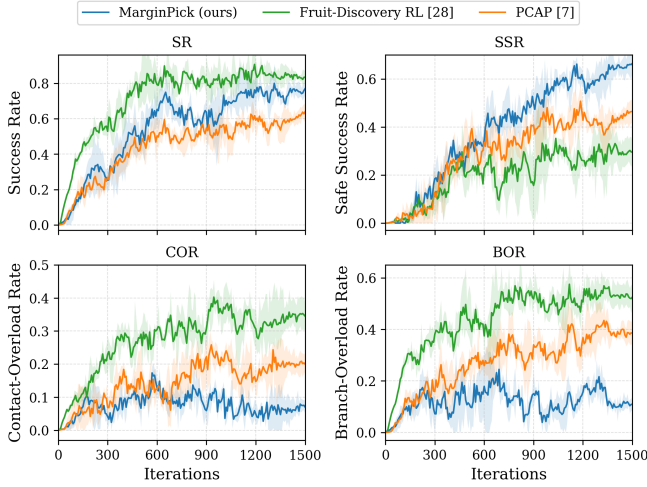


Fig. 3. Training curves for the benchmark policies over 1,500 PPO iterations. Panels report SR, SSR, COR, and BOR; curves show five-seed means with min-to-max shaded ranges.

Evaluation is performed on held-out tree scenes that are not used for policy training; all methods are tested on the same scene seeds and under the same 6-D Cartesian action interface.

All training stages use 4096 parallel environments and are run on a workstation with an Intel Core i5-14600KF CPU, 32 GB RAM, and an NVIDIA GeForce RTX 4070 SUPER GPU with 12 GB memory. The imitation prior is trained from DLS-IK demonstrations collected in parallel simulation. During demonstration collection, the simulator is run for 300 iterations to obtain a large set of IK-labeled state-action pairs. BC training uses 1000 supervised updates. The DAgger refinement is performed for 10 data-aggregation rounds. In each round, the latest BC policy is loaded into the vectorized simulator and rolled out for 10 collection iterations; all visited states are relabeled by the DLS-IK demonstrator and merged into the aggregated imitation dataset. The policy is then trained for 50 supervised updates on the aggregated dataset, and the resulting model is used to initialize the next data-aggregation round. Residual PPO then fine-tunes the policy for 1500 iterations and 16 steps per environment.

Reward weights and utilization-aggregation coefficients in Sec. IV-B are fixed across training runs unless explicitly ablated; the compact configuration is listed in Table I.

B. Metrics

We report four episode-level rates:

- 1) **Success Rate (SR)**: the fraction of trials in which the end-effector reaches the pre-harvest band with the required approach orientation.
- 2) **Safe Success Rate (SSR)**: the fraction of trials that satisfy the SR condition while keeping all robot-link contact forces and branch deflections within their prescribed safety limits throughout the rollout.
- 3) **Contact-Overload Rate (COR)**: the fraction of trials in which at least one arm link reaches or exceeds its force utilization limit.

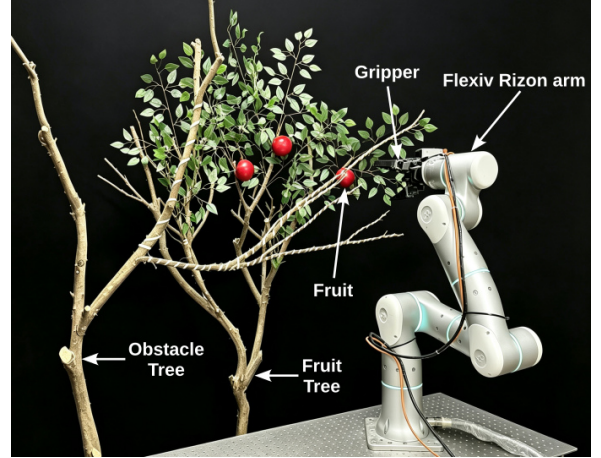


Fig. 4. Real-robot validation scene: a Flexiv Rizon arm reaches red fruit proxies on the target tree through clutter created by a separate obstacle tree.

- 4) **Branch-Overload Rate (BOR)**: the fraction of trials in which any fruit-tree or obstacle-tree branch exceeds its branch utilization limit.

This separation is important because a policy can achieve a high SR by reaching the target after forcing its way through branches; such episodes should improve SR but not SSR once contact force or branch deflection breaches the prescribed safety limits, and they should also contribute to the COR or BOR event rates.

C. Ablation Study

The ablation study compares two complementary variant groups against the complete MarginReach policy reported as the final reference row. The first group evaluates the learning framework: (1) *BC Prior* executes the frozen behavior-cloned reaching prior without reinforcement learning; (2) *Vanilla PPO* trains the actor from scratch with the same task observation and reward but without imitation initialization or residual decomposition; and (3) *BC-Initialized PPO* initializes the PPO actor from the behavior-cloned policy and then continues standard PPO fine-tuning. The second group evaluates the margin utilization mechanism within the residual framework: (1) *w/o contact observation* removes link-wise contact utilization feedback from the policy input; (2) *w/o contact penalty* removes the graded contact utilization cost while preserving branch penalties; (3) *w/o branch penalty* removes fruit-tree and obstacle-tree branch utilization costs while preserving contact penalties; and (4) *w/o margin penalties* removes both contact and branch utilization penalty terms. These ablation variants are evaluated with the same metrics as the benchmark in Table II.

Table II shows that the proposed margin-aware residual formulation improves safe reaching rather than simply maximizing nominal completion. The BC prior achieves the highest SR, but its low SSR and high COR/BOR indicate frequent mechanically unsafe interactions in randomized branch clutter. PPO-based variants improve adaptability, but still exhibit

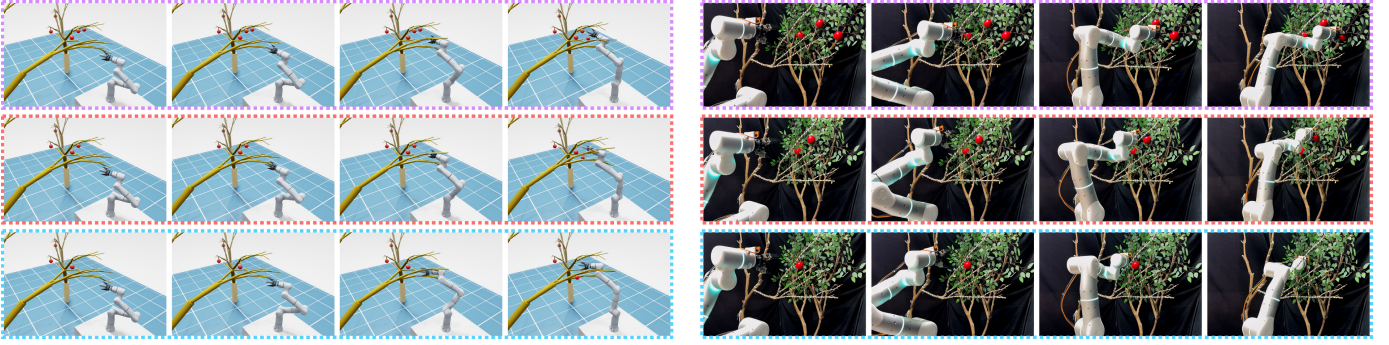


Fig. 5. Representative rollout sequences from the simulation and real-robot evaluations, showing the arm navigating branch clutter toward the pre-harvest target.

TABLE II
ABLATION STUDY.

Variant	SR (%)	SSR (%)	COR (%)	BOR (%)
<i>Learning framework</i>				
(1) BC Prior	92.4	26.7	54.1	61.8
(2) Vanilla PPO	79.2	39.5	43.7	52.3
(3) BC-Initialized PPO	81.6	42.8	40.4	51.9
<i>Margin utilization</i>				
(1) w/o contact observation	72.1	53.6	20.8	27.4
(2) w/o contact penalty	67.9	42.2	26.5	19.7
(3) w/o branch penalty	72.3	59.1	11.6	26.2
(4) w/o margin penalties	86.8	31.4	46.9	53.5
MarginReach	79.3	63.5	9.8	14.4

substantial overload without the complete residual margin structure.

The margin ablations isolate the role of each safety signal. Removing contact-margin observation lowers SSR, showing that force-utilization feedback is needed for reactive motion around branch contact. Removing the contact penalty mainly increases COR, whereas removing the branch-utilization penalty increases BOR. Without both margin penalties, the policy keeps relatively high SR but suffers a large SR–SSR gap. The full MarginReach achieves the highest SSR and the lowest COR/BOR, indicating that the residual policy uses contact and branch utilization to balance reaching progress against overload avoidance.

D. Benchmark Comparisons

We compare MarginReach with two literature-derived policies that address the closest neighboring problems in contact-rich canopy interaction. *PCAP* [7] is a proprioceptive contact-aware reaching policy originally designed for gentle motion through tree branches; in our benchmark it uses the same reaching target and Cartesian controller, with contact represented by binary whole-arm contact cues. *Fruit-Discovery RL* [26] is a reinforcement-learning policy that incorporates fruit-discovery guidance during canopy interaction; in our benchmark this guidance is implemented with a simulator-derived fruit-visibility signal.

All methods are evaluated with the same robot, generated tree scenes, 6-D Cartesian action interface, DLS-IK controller, held-out evaluation seeds, and episode-level metrics. The main comparison is reported in Table III; Fig. 3 shows the corresponding training trends for success and overload rates.

TABLE III
BENCHMARK COMPARISONS.

Method	SR (%)	SSR (%)	COR (%)	BOR (%)
<i>Simulation</i>				
PCAP [7]	73.6	54.2	14.9	33.1
Fruit-Disc. RL [26]	83.4	36.8	27.5	45.7
MarginReach	79.3	63.5	9.8	14.4
<i>Real</i>				
PCAP [7]	73.3	50.0	16.7	33.3
Fruit-Disc. RL [26]	66.7	33.3	30.0	43.3
MarginReach	76.7	60.0	13.3	23.3

The simulation results in Table III show that MarginReach achieves the best safe-success performance among the compared methods. Fruit-Discovery RL obtains the highest SR, but its lower SSR and higher overload rates indicate that target-directed guidance alone can encourage aggressive canopy interaction. PCAP reduces contact overload compared with Fruit-Discovery RL, but its binary contact representation lacks graded margin information for branch safety. In contrast, MarginReach gives up a small amount of nominal SR relative to the most success-oriented baseline while achieving the highest SSR and the lowest COR/BOR, supporting the benefit of normalized force and branch-deflection utilization for balancing reaching success against mechanical safety.

E. Real-World Validation and Failure Analysis

The physical validation setup is shown in Fig. 4. A Flexiv Rizon arm reaches toward red fruit proxies mounted on a fruit tree, while an independently placed obstacle tree introduces branch clutter between the robot and the target. This arrangement mirrors the simulated evaluation scene but uses real compliant branches to test pre-harvest reaching under physical interference. The trained policy is kept fixed with the same observation and 6-D Cartesian action interface as simulation; each command is interpolated, converted by the DLS-IK layer, and low-pass filtered. Each method is evaluated over 30 physical trials, with aggregated results reported in Table III.

Following force-aware branch manipulation practice [27], hardware contact load is estimated from the external joint-torque residual after the robot controller’s gravity and dynamics compensation. The residual torque is mapped to an apparent Cartesian contact wrench through the damped inverse-transpose Jacobian of the active contact frame, and its force

magnitude is low-pass filtered before thresholding. For branch interaction, the hardware evaluation uses an operator safety-stop protocol. A single operator applied the same predefined visual stop criterion across all methods and trials. The criterion was triggered when continued motion was judged to risk permanent branch damage, such as visible cracking, sharp localized bending near the branch base, or rapidly increasing deformation under sustained contact. Such trials were terminated and counted as branch-limit events. The physical comparison uses the same policies and task-level metrics as Sec. V-D over matched fruit positions and obstacle-tree configurations. SR and SSR follow the simulation definitions, while COR and BOR are computed from estimated contact-load overloads and branch-limit safety stops, respectively. Fig. 5 summarizes representative simulation and real-robot rollouts.

The real-robot results show the same qualitative ordering as the simulation results: MarginReach achieves the highest SR and SSR while reducing both contact- and branch-overload occurrence. The remaining SR–SSR gap reflects physical sensing and branch-compliance variation, while the lower COR and BOR indicate that the margin-aware policy retains lower-overload behavior on hardware.

VI. CONCLUSION

This paper studied pre-harvest fruit reaching in dense, deformable canopies, where lower-risk execution requires regulated interaction rather than strict contact avoidance. We introduced MarginReach, a margin-aware residual RL framework that converts link contact force and branch deflection into utilization signals and uses them to guide Cartesian residual corrections on top of a behavior-cloned reaching prior. Together with the procedural Isaac Lab tree environment, this formulation enables policies to exploit mild branch compliance while discouraging force and deflection overloads.

The current study focuses on reaching a pre-harvest standoff pose as the contact-rich precursor to grasping and detachment. Future work will extend the policy to full harvest execution, further improve plant-mechanics calibration, and incorporate richer sensing for contact and branch-state estimation in physical canopies.

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